



Empowering Scientific Workflows with Federated Agents

Alok Kamatar, J. Gregory Pauloski, Yadu Babuji, Ryan Chard, Mansi Sakarvadia, Daniel Babnigg, Kyle Chard, Ian Foster, Suman Raj, Augustus Ellerm



Argonne
NATIONAL
LABORATORY

Motivation

- Scientific workflows are growing more complex, spanning HPC systems, experimental facilities, and data repositories.
- Humans still oversee most core decisions, slowing discovery. Intelligent agents can perform tasks to automate the research cycle.
- Current agentic frameworks are centralized, cloud-native, and designed for conversational AI, not for federated HPC infrastructure with diverse access protocols and variable availability.
- New middleware is needed to seamlessly deploy agents across federated research infrastructure: supercomputers, instruments, and data repositories.

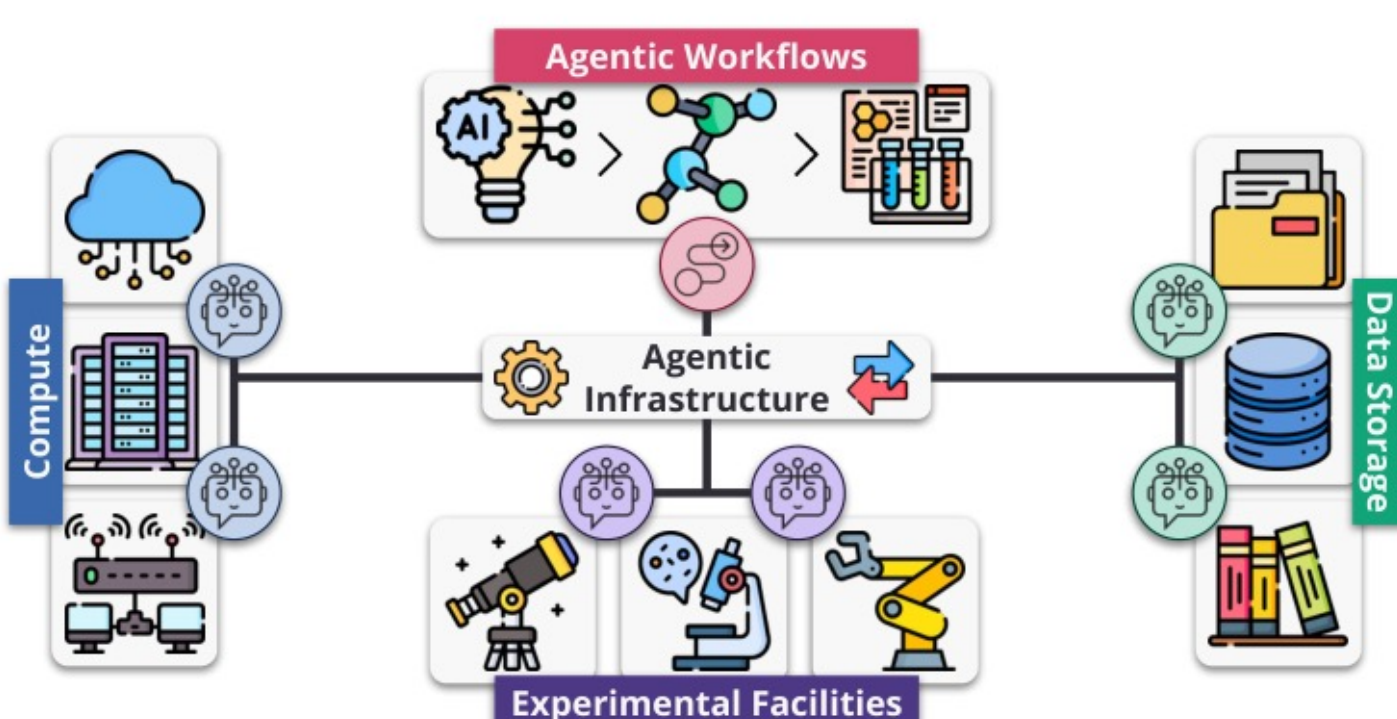


Fig. 1 Cooperative agents spanning federated research infrastructure (HPC, instruments, data storage) enable agentic workflows.

Requirements for Federated Agents

R1 – Federated Orchestration

Launch and manage agents across different computing systems and scientific instruments.

R2 – Configurable Data Plane

Support HPC high-speed interconnects, large-volume transfers, and high-frequency communication at simulation time-step rates.

R3 – Temporally Decoupled Messaging

Handle variable resource availability; research infrastructure has much lower uptime than cloud infrastructure.

R4 – Agent Authentication & Permissions

Securely delegate (and revoke) per-agent permissions to use research infrastructure via Globus Auth with standard TLS and access control.

R5 – Resilient State Management

Maintain stateful agent properties despite dynamic resource conditions and scheduler interruptions.

Background: Use Cases

Steering Applications

Agents observe and learn from prior results to steer simulation campaigns adaptively (e.g., Colmena for molecule discovery; agentic systems for on-the-fly model switching in physical simulations).

Decomposing Applications across Facilities

Multi-site workflows distribute heterogeneous computations across HPC clusters (e.g., MOFA: generative AI on Aurora, simulations on Polaris, screening on Cloud).

Integrating Instruments into Workflows

Embodied agents co-located with scientific instruments (e.g., telescopes) track state, calibrate data in real time, and feed results back into the workflow pipeline.

Conversational Research Assistants

Multi-agent LLM pipelines autonomously parse, query, and validate information from scientific publications to accelerate cross-disciplinary insights.



Upcoming Talks!

- IPDPS 5/26/2026 New Orleans
- PEARC 7/27/2026 Minneapolis
- TPC 5/31/2026 Baltimore
- ISC 6/22/2026 Germany

ACADEMY Architecture

Modular Python middleware (actor-based) for building agentic workflows across federated research infrastructure.

Agents

@action methods handle concurrent async requests; @loop decorators run background control loops; agent_on_startup / agent_on_shutdown callbacks manage lifecycle. Multiple-inheritance enables composite agents.

Handles

Client interface to a remote agent. Translates each method call into an async request/response message via the Exchange. Handles auto-attach to the caller's mailbox, ensuring one listener per client.

Exchange (Data Plane)

- Thread Exchange – in-memory; single-process testing.
- Hybrid Exchange – TCP direct messaging + Redis object store; ideal for HPC node-local networks; multi-path design tolerates transient failures.
- Cloud Exchange – REST API over Redis with Globus Auth; cross-facility federated deployments with TLS security.

Executors (Control Plane)

Pluggable launchers: Parsl High-throughput Executor, ProcessPool, ThreadPool. Parsl enables dynamic scaling of agent resource pools based on workload.

Manager

Combines an Exchange with one or more Executors. Provides a single interface for launching, using, and cleaning up agents. Reduces boilerplate.

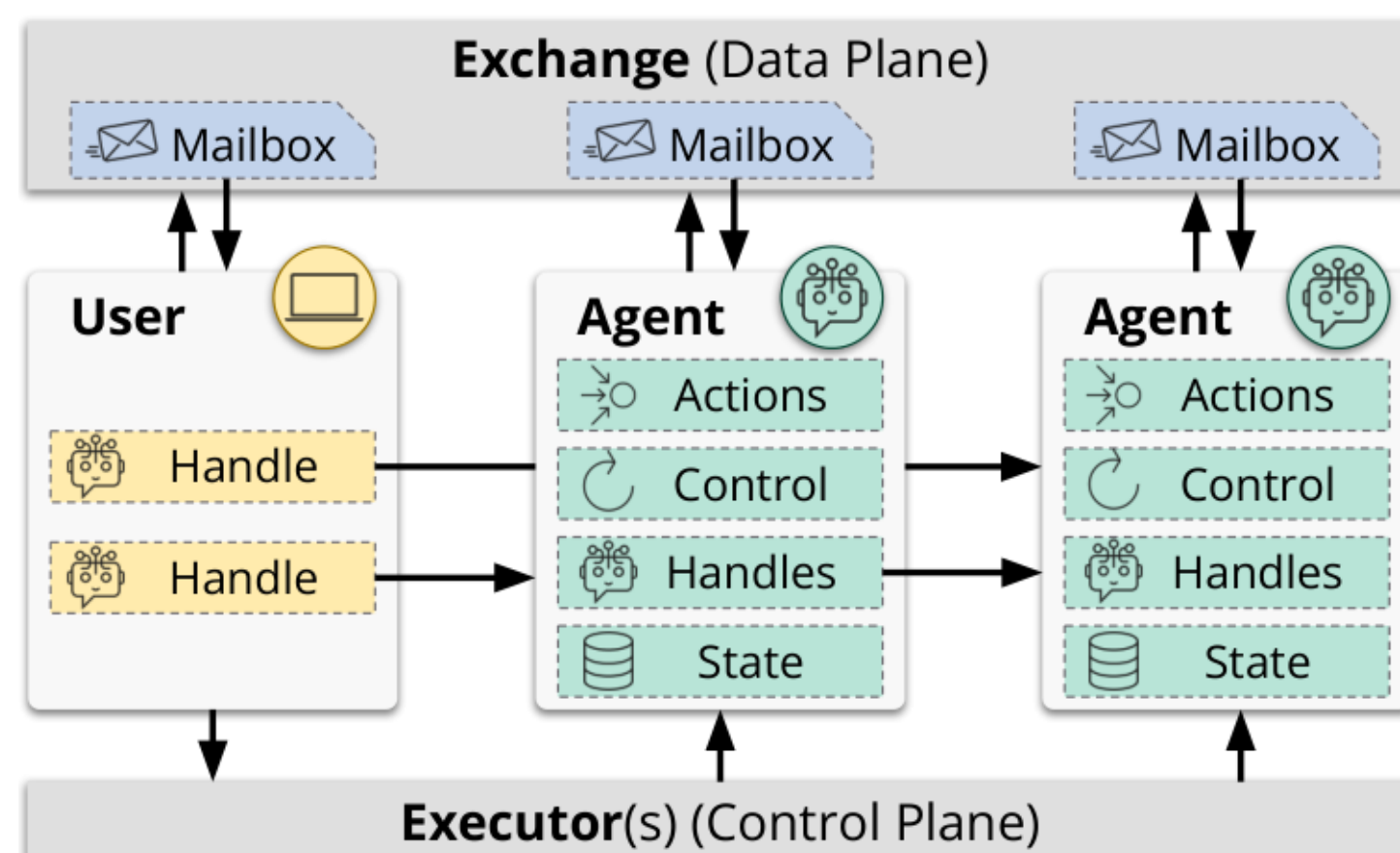


Fig. 2 ACADEMY architecture: agents/users interact via handles; Exchange (data plane) and Executors (control plane) are decoupled.

Case Studies

Materials Discovery – MOFA

Online learning pipeline generating and screening Metal-Organic Frameworks (MOFs) for CO₂ capture. Five-stage workflow deployed across Aurora, Polaris, and Chameleon Cloud. (Refer Fig.3)

Astronomical Spectroscopy

Integral-field spectrograph data calibration at Keck Observatory. An ACADEMY agent tracks telescope optical parameters over time, using previously calibrated frames to accelerate alignment and reduce noise in data cubes.

Decentralized Machine Learning

1,500+ autonomous agents on Aurora GPU tiles collaboratively train MNIST models using peer-to-peer communication (power-law connectivity graph, ProxyStore pass-by-ref backend). No central coordinator required.

Information Extraction

Multi-agent pipeline parses 10 scientific PDFs and generates, validates, and scores multiple-choice questions using Mistral-7B and Llama-3-70B agents.

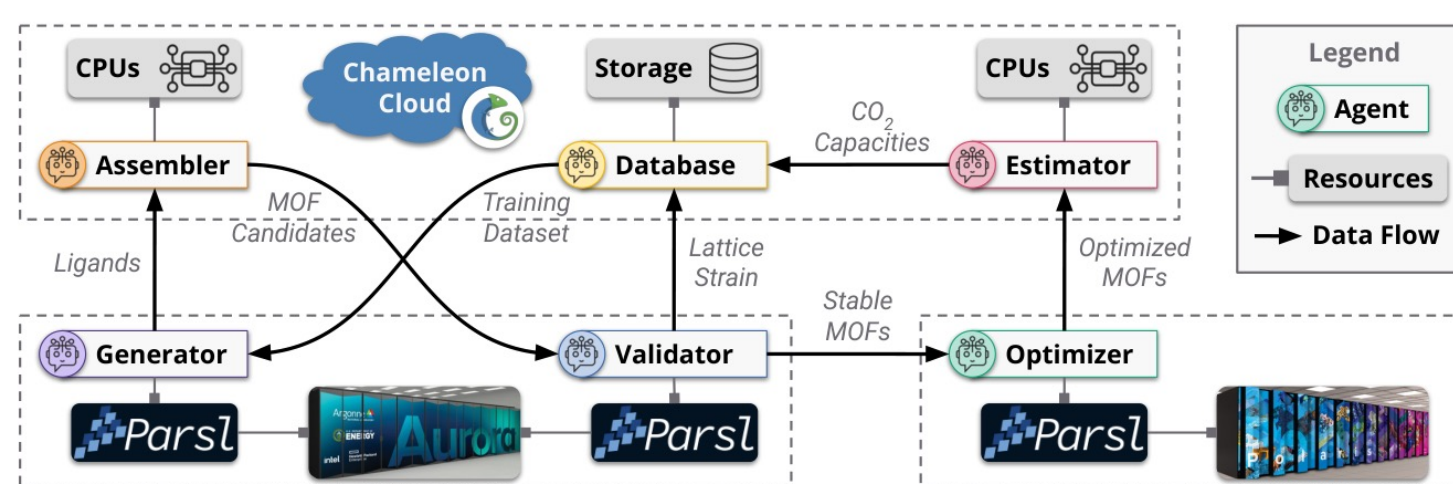


Fig. 3 MOFA deploys agents across Aurora, Polaris, and Chameleon Cloud using Globus Compute for federated task execution.

Performance Evaluation

Weak Scaling – Agent Startup

- Aurora nodes: AMD EPYC Genoa CPUs, 52 cores, Intel Data Center Max GPUs.
- Warm-start: ACADEMY launches a single agent in 5.5 ms, 2.8× faster than Dask.
- Scales to 3,328 agents in 7.6 s vs. Dask's 23.4 s on Aurora (ALCF).

Action Latency

- Intra-site (single Aurora node): 385±301 μs vs. 1,186±1,059 μs (Dask) and 526±308 μs (Ray).
- Inter-site (workstation → Aurora): ACADEMY 34±5 ms vs. Globus Compute 1,226±313 ms for stateful 1 GB payloads.
- Pass-by-ref with ProxyStore reduces 100 MB payload latency by 97.6% (intra-site) and 45.1% (inter-site).

Agent Messaging Throughput

- Agent-to-agent chat (ACADEMY vs AutoGen distributed runtime): ACADEMY consistently lower completion time across all message sizes.

Memory Efficiency

- ACADEMY+Parsl is the most memory-efficient option at every agent count tested (up to 104 agents on a single Aurora node).

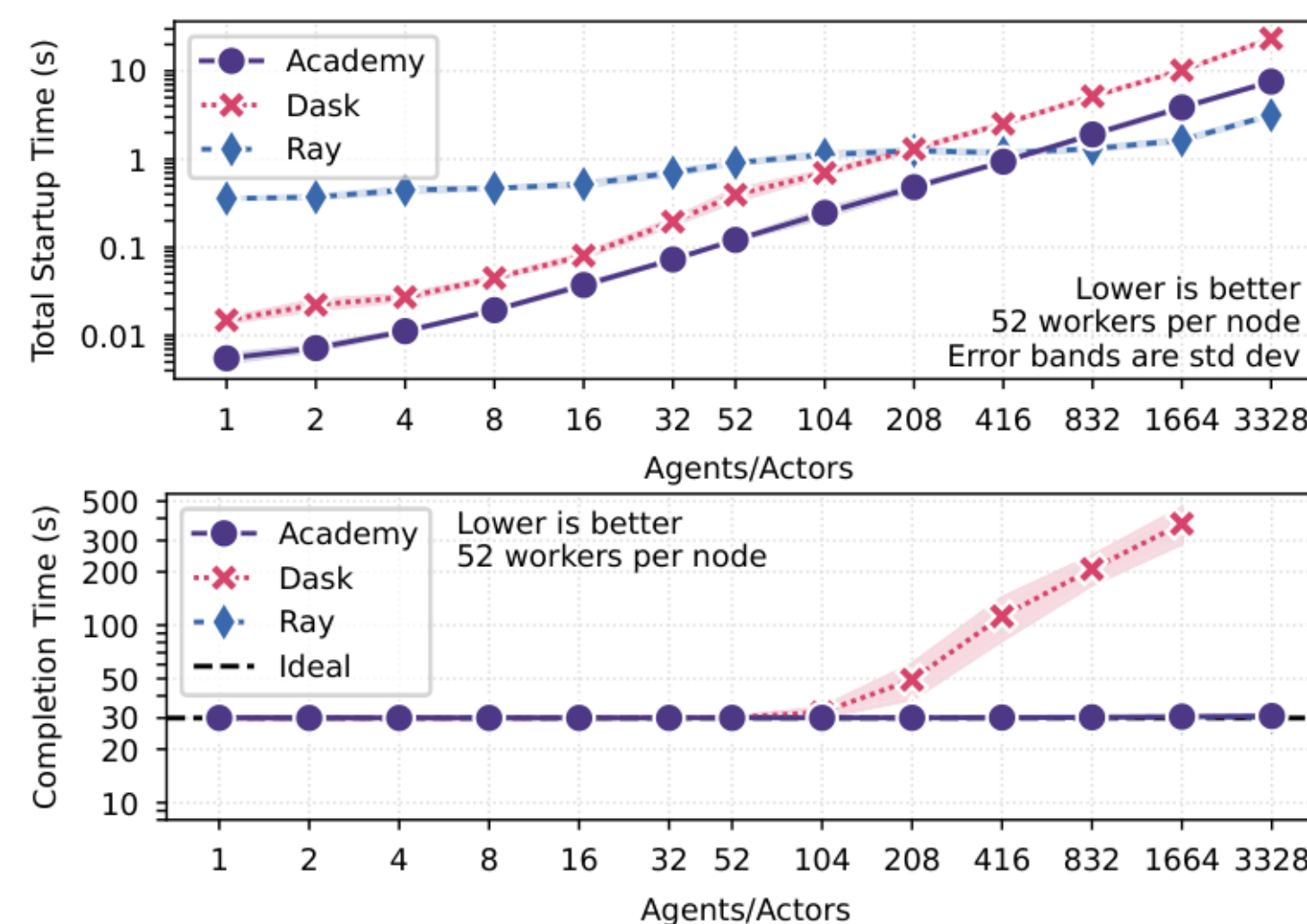


Fig. 4 Weak scaling: warm-start time (top) and action completion time (bottom) for ACADEMY, Dask, and Ray up to 3,328 agents.

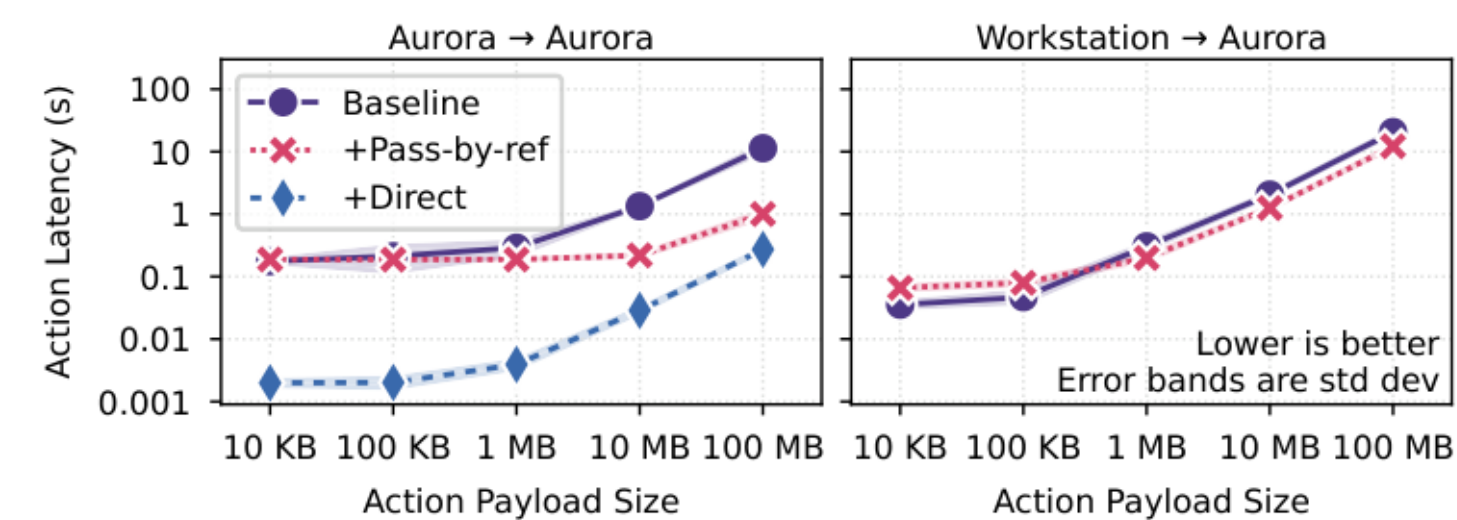


Fig. 5 Client invocation latency vs. payload size with different optimizations on the hybrid and cloud exchanges.

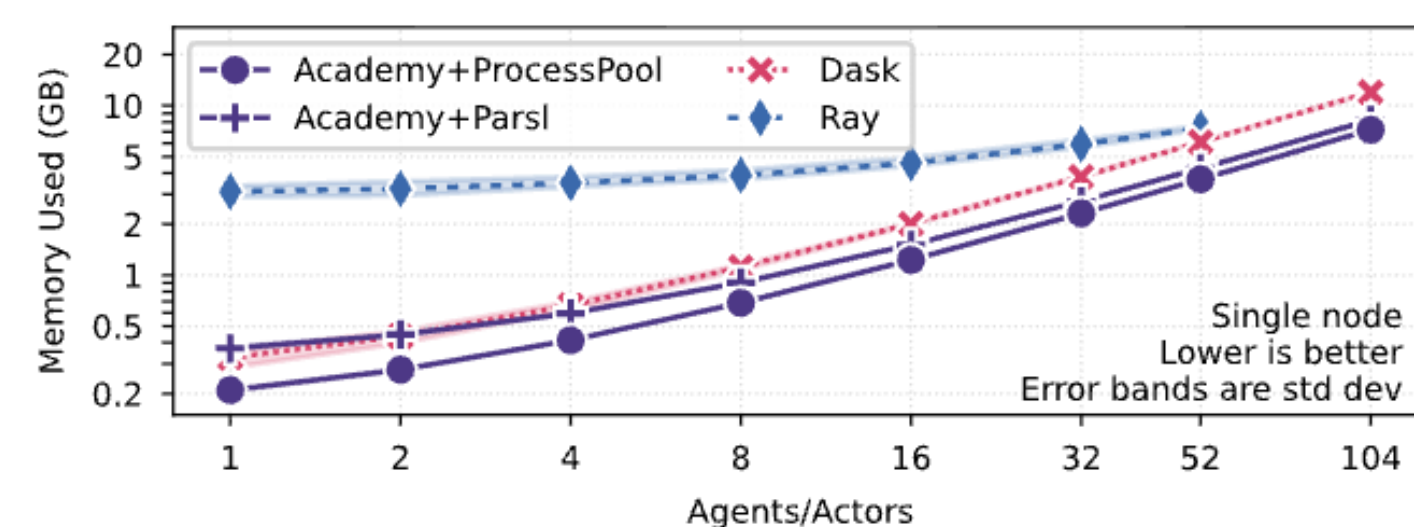


Fig. 6 Memory used by n agents/actors

Putting It Together

ACADEMY enables scalable, flexible orchestration of intelligent agents across heterogeneous HPC resources, addressing three key challenges:

- Representing and programming agents
- Communicating among agents
- Executing agents across federated resources

Microbenchmarks confirm high performance competitive with Dask and Ray.

Four case studies spanning materials science, astronomy, decentralized learning, and information extraction validate the breadth and practicality of ACADEMY.

Acknowledgements

